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Assignment 7A

Aim
Write a menu driven program to implement stack using linked list.

Theory
Stack is a linear data structure which is implemented using array or linked list.

Push (Insert)
It adds an element to the start of the linked list (stack). The node of the element points to the element at the start ~~of~~ in the linked list.

POP (Delete)
It removes an element at the top of the stack which is the first element of the linked list and the node is freed.

PEEK (Top element)
It displays the top element of the stack that is the first element of the linked list

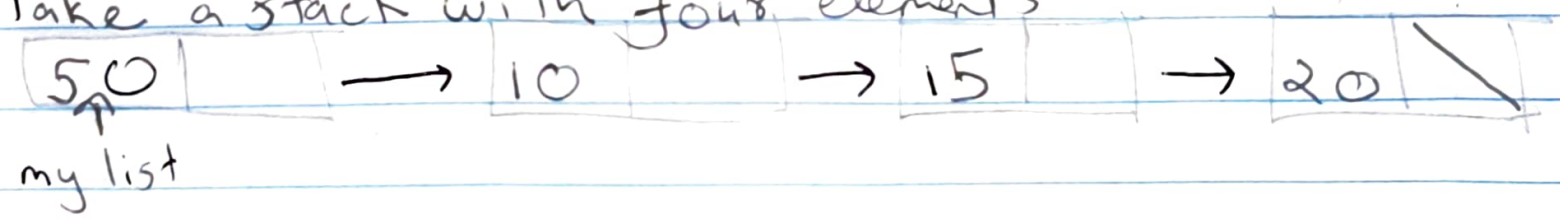
Time complexity
The time complexity is O(n) as all elements need to be traversed.

Conclusion

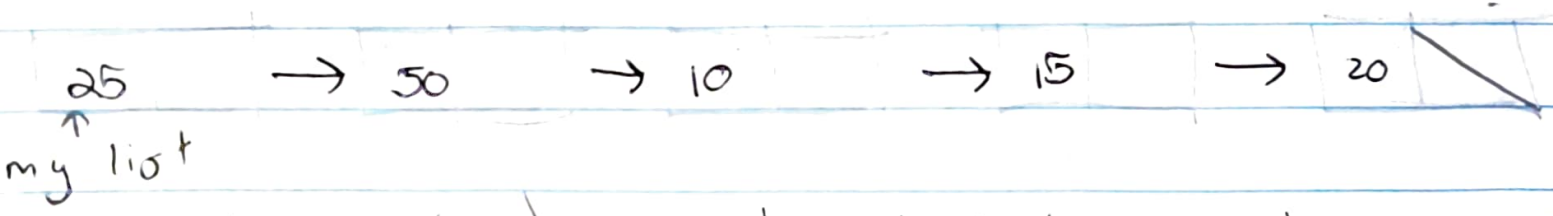
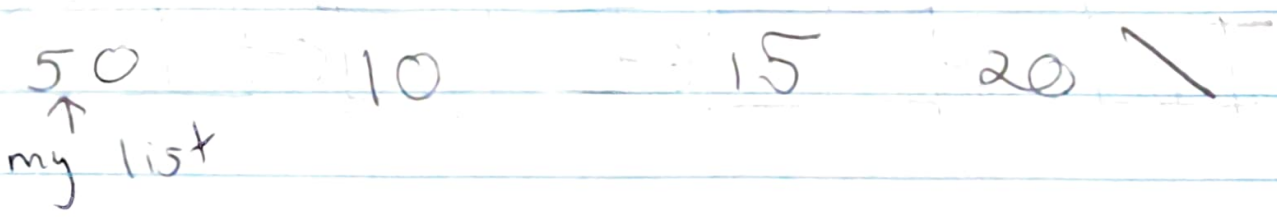
The concept of linked lists is applied to evaluate problems in a variety of applications.

Working Example

Take a stack with four elements



Now to push an element (25) insert it in the beginning and point the node to the first



Now to pop (25), point my list to 50 and free 25